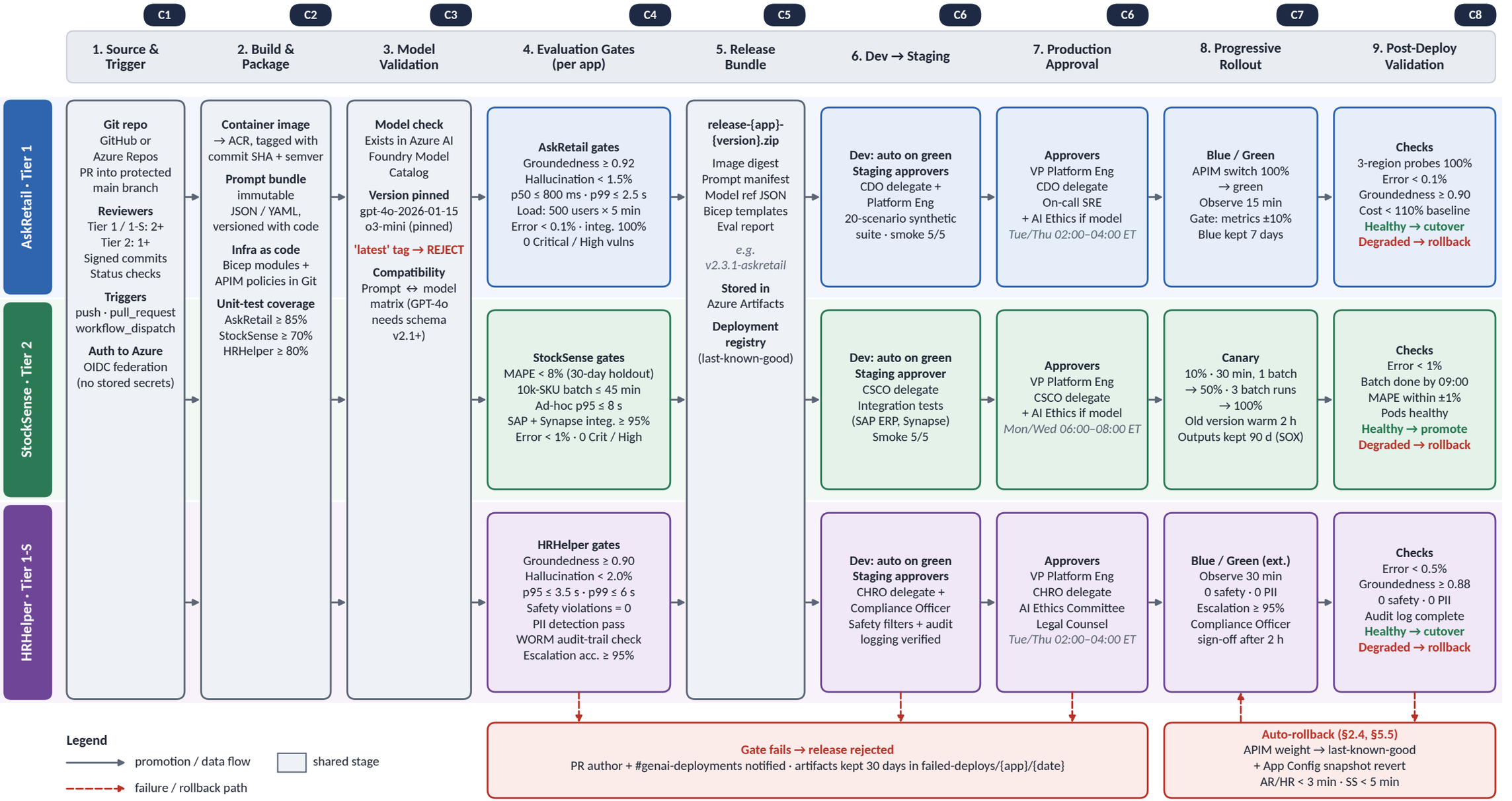
# Diagram Set — Callout Map

Companion to Dennis_EnterpriseDeployment_Package. Each numbered callout on Figures 1–3 maps to the document section that explains it. Figures follow on pages 2–4.

| Callout | Figure | Diagram element | Document section | What it shows |
|---|---|---|---|---|
| C1 | Fig. 1 | **Source & trigger** | §2.1 | Protected main; 2+ reviewers Tier 1, 1+ Tier 2; signed commits; OIDC, no stored secrets |
| C2 | Fig. 1 | **Build & package** | §2.1 | Images tagged SHA + semver, deployed by digest; immutable prompt bundles; Bicep/APIM policy in Git |
| C3 | Fig. 1 | **Model validation** | §2.1 | Versions pinned in Foundry Model Catalog; 'latest' rejected; prompt–model compatibility check |
| C4 | Fig. 1 | **Evaluation gates (per app)** | §2.2 | Numeric quality, latency, test, security and cost gates for each app |
| C5 | Fig. 1 | **Release bundle** | §2.1 | release-{app}-{version}.zip; deployment registry tracks last-known-good |
| C6 | Fig. 1 | **Dev → staging → production approval** | §2.3 | Tiered approvers; AI Ethics on every model promotion; CFO above $50K/month |
| C7 | Fig. 1 | **Progressive rollout** | §3.2, §5.3 | Blue/green (AskRetail); canary 10→50→100% (StockSense); extended blue/green (HRHelper) |
| C8 | Fig. 1 | **Post-deploy validation, auto-rollback** | §2.4, §5.5 | Automated triggers; APIM weight + App Configuration snapshot revert |
| C9 | Fig. 2 | **APIM: authenticate** | §3.1 | Entra ID tokens; per-app subscriptions; managed identity to Azure OpenAI |
| C10 | Fig. 2 | **APIM: weighted routing** | §3.1, §3.2 | Weighted pools per version; cross-region failover; GPT-4o-mini fallback |
| C11 | Fig. 2 | **Model gateway: token quotas** | §3.1 | Daily budgets 4.5M / 1.2M / 0.8M tokens; CostCenter-tagged showback |
| C12 | Fig. 2 | **APIM: API versioning** | §3.1 | Path versions for breaking changes; revisions otherwise; 60-day deprecation |
| C13 | Fig. 2 | **Orchestrators** | §3.3 | AKS (AskRetail); Azure Container Apps (StockSense); private AKS (HRHelper) |
| C14 | Fig. 2 | **Release control plane** | §3.3 | Per-release App Configuration snapshot; one manifest for every orchestrator |
| C15 | Fig. 3 | **Trace capture points T1–T6** | §4.1 | W3C Trace Context; OpenTelemetry → Application Insights |
| C16 | Fig. 3 | **LLM-specific signals** | §4.2 | Groundedness, hallucination, drift, tokens/cost, safety and PII thresholds |
| C17 | Fig. 3 | **Continuous evaluation loop** | §4.3 | 2–5% replay against candidates; drift blocks promotion |
| C18 | Fig. 3 | **Alert routing** | §4.4 | P0–P4 mapped to SEV-0–SEV-3 with response times and routes |
| C19 | Fig. 3 | **Dashboards** | §4.5 | Managed Grafana (platform), Workbooks (app owners), Power BI (executive/CFO) |
| C20 | Fig. 3 | **Runbook hand-off** | §5.4, §5.5 | SEV-0/1 → incident response → rollback execution |

# Figure 1 — CI/CD Pipeline Architecture (NRC GenAIOps Platform)

Grey stages are shared and run once per release; evaluation, promotion and rollout run in parallel per-app lanes. Red dashed paths are failure paths.

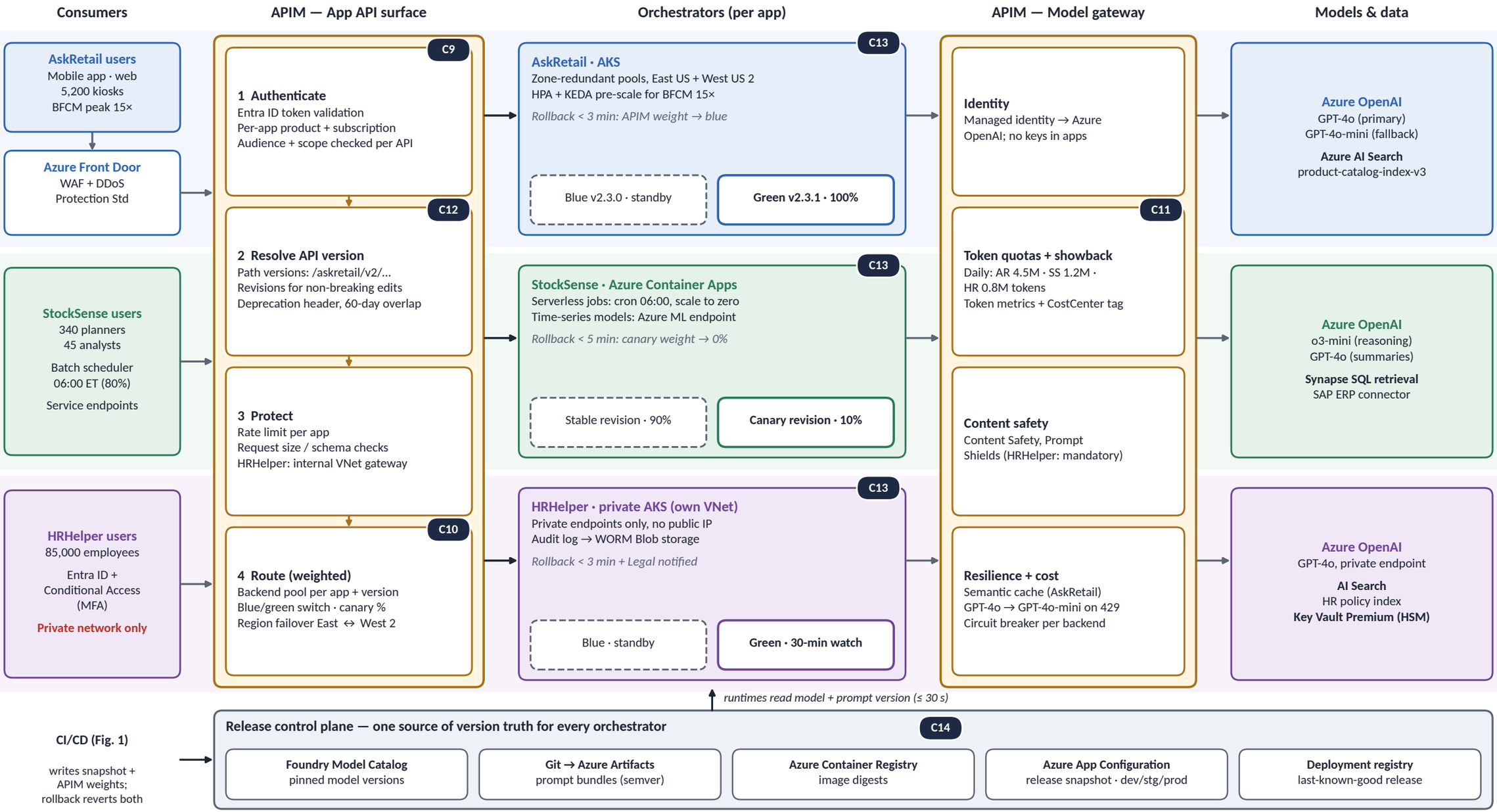



**Callout key (callout → document section)**

- **C1** Source, branch protection, triggers → §2.1
- **C2** Build; versioned images, prompts, IaC → §2.1
- **C3** Model validation; version pinning → §2.1
- **C4** Per-app automated evaluation gates → §2.2
- **C5** Release bundle + deployment registry → §2.1
- **C6** Environment promotion and approvals → §2.3
- **C7** Progressive rollout per app → §3.2, §5.3
- **C8** Post-deploy validation, rollback → §2.4, §5.5

# Figure 2 — AI Gateway and Deployment Operations

Both amber columns are the same Azure API Management Premium instance (East US primary, West US 2 secondary): app APIs for clients, and a model gateway that every LLM call passes through.




**Callout key (callout → document section)**

- C9 Gateway authentication → §3.1
- C10 Weighted routing, failover → §3.1, §3.2
- C11 Token quotas and showback → §3.1
- C12 API versioning → §3.1
- C13 Orchestrator choice per app → §3.3
- C14 Model + prompt versioning → §3.3

# Figure 3 — Observability Architecture

Traces are captured at every hop, stored in Azure Monitor, scored for LLM quality by a continuous evaluation loop, and turned into severity-routed alerts and role-based dashboards.

**Request path — one trace per conversation turn; W3C traceparent propagated through every hop** C15

**T1 Gateway span**
*APIM*
auth · rate limit · route
http.response.status_code

→

**T2 Orchestrator span**
*AKS / Container Apps*
prompt load (App Config)
ai.application.version

→

**T3 Retrieval span**
*Azure AI Search*
vector search + rerank
ai.retrieval.source

→

**T4 Model inference span**
*Azure OpenAI via APIM*
ai.model.name
prompt / completion tokens

→

**T5 Safety + quality span**
*Content Safety · Presidio*
groundedness score
hallucination flag · PII

→

**T6 Audit span**
*Log Analytics · WORM Blob*
ai.conversation.id
immutable for HRHelper

**Common span attributes (OpenTelemetry): ai.application.name / version · ai.model.name (pinned, no 'latest') · ai.total.tokens · ai.groundedness.score · ai.hallucination.flag · ai.user.tier · cloud.region**

*spans, logs, metrics* ↓

## Telemetry pipeline and stores

**OpenTelemetry SDK → Collector → Azure Monitor exporter**

**Application Insights — traces**
100% of Tier 1 errors, 10% of successes · 90 d

**Log Analytics (KQL) — logs**
90 d hot, 1 y archive · runbook KQL queries

**Azure Monitor Metrics + Prometheus**
15 s infrastructure · 60 s business metrics

**APIM token metrics + Cost Mgmt**
showback by CostCenter tag · budget alerts

*sampled live traffic* ↓

## Continuous evaluation loop C17

**Event Hub sampler**
2–5% of live traffic

→

**Replay runner**
AKS job: current vs candidate

→

**AI Evaluation SDK**
groundedness, relevance, safety

↓

**Cosmos DB**
evaluation results

←

**Drift engine**
KS > 0.15 or W > 0.3

←

**Drift → block promotion**

**No drift → new baseline**

**AskRetail: 2% (≥ 500/day), every 4 h · block if ≥ 2 metrics fall > 5%**
**StockSense: 5% of batch jobs, after each run · block if MAPE rises > 1%**
**HRHelper: 3% (≥ 100/day), every 6 h · block if any safety metric falls**

*scores on every span* ↓

## LLM-specific signals and thresholds C16

| | |
|---|---|
| **Groundedness** | AR ≥ 0.92 · HR ≥ 0.90 |
| **Hallucination rate** | AR < 1.5% · HR < 2.0% |
| **Semantic drift** | KS > 0.15 or W > 0.3 for 2 h |
| **Relevance / coherence** | ≥ 0.85 / ≥ 0.80 (all apps) |
| **Cost per conversation** | ≤ $0.04 / $0.12 / $0.08 |
| **Quota utilisation** | < 85% / < 90% / < 80% |
| **Content safety** | severity ≥ 2 blocked |
| **PII leakage** | 0 incidents (any → P0) |
| **Forecast (StockSense)** | MAPE < 8% · batch ≤ 45 min |

*Order of three values: AskRetail / StockSense / HRHelper*

## Alert routing — Azure Monitor alert rules → Action Groups → runbook severity C18

| | | | |
|---|---|---|---|
| **P0** | **SEV-0 · < 5 min** | PagerDuty: SRE + VP Platform (+ CISO) | *outage, PII leak* |
| **P1** | **SEV-1 · < 15 min** | PagerDuty: on-call SRE + App Owner | *hallucination > 3%, safety ≥ 4* |
| **P2** | **SEV-2 · < 1 h** | Teams + ticket | *groundedness −0.05, drift, cost* |
| **P3** | **SEV-3 · < 4 h** | Email digest + next-day ticket | *cache < 80%, quota > 75%* |
| **P4** | **Info** | Dashboard / weekly report | *deploy succeeded* |

**SEV-0 / SEV-1**
runbook §5.4
→ rollback §5.5

C20

## Dashboards by audience C19

**Platform Ops**
*Azure Managed Grafana*
SRE · VP Platform · CFO del.
all-app health (RAG)
latency heatmap, errors
tokens vs quota, cost/h
alerts, last 10 releases

**App Owner**
*Azure Monitor Workbooks*
CDO · CSCO · CHRO del.
AR: groundedness, halluc.
SS: MAPE, batch timeline
HR: escalation, safety (zero-tolerance), audit

**Executive / CFO**
*Power BI*
CFO · CDO · Board
spend vs $1.4M/month
cost per outcome
ROI by application
compliance posture

### Callout key (callout → document section)

- C15 Trace capture points T1–T6 → §4.1
- C16 LLM-specific metrics → §4.2
- C17 Continuous evaluation → §4.3
- C18 Alert-to-severity mapping → §4.4
- C19 Dashboards → §4.5
- C20 Hand-off to runbook → §5.4, §5.5